Table 2.2 Binary, Octal, Hexadecimal equivalent of Decimal Numbers

| Decimal | Binary | Octal | Hexadecimal |
|---|---|---|---|
| 0 | 0000 | 000 | 0000 |
| 1 | 0001 | 001 | 0001 |
| 2 | 0010 | 002 | 0002 |
| 3 | 0011 | 003 | 0003 |
| 4 | 0100 | 004 | 0004 |
| 5 | 0101 | 005 | 0005 |
| 6 | 0110 | 006 | 0006 |
| 7 | 0111 | 007 | 0007 |
| 8 | 1000 | 010 | 0008 |
| 9 | 1001 | 011 | 0009 |
| 10 | 1010 | 012 | A |
| 11 | 1011 | 013 | B |
| 12 | 1100 | 014 | C |
| 13 | 1101 | 015 | D |
| 14 | 1110 | 016 | E |
| 15 | 1111 | 017 | F |

**Example**

The hexadecimal sequence $(25)_{16}$ has the decimal equivalent:

$$(25)_{16} = 2\times16^1 + 5\times16^0$$
$$= 32+5$$
$$= (37)_{10}$$

**Workshop**

1. Identify the number system for the following numbers

| S. No. | Number | Number system |
|---|---|---|
| 1 | $(1010)_{10}$ | Decimal Number system |
| 2 | $(1010)_{2}$ | |
| 3 | $(989)_{16}$ | |
| 4 | $(750)_{8}$ | |
| 5 | $(926)_{10}$ | |

2. State whether the following numbers are valid or not. If invalid, give reason.

| S.No. | Statement | Yes / No | Reason (If invalid) |
|---|---|---|---|
| 1. | 786 is an Octal number | | |
| 2. | 101 is a Binary number | | |
| 3. | Radix of Octal number is 7 | | |

## 2.4 Number System Conversions

### 2.4.1 Decimal to Binary Conversion

Generally two methods followed.

**Method 1:** To convert Decimal to Binary "**Repeated Division by 2**" method can be used. Any Decimal number divided by 2 will leave a remainder of 0 or 1. Repeated division by 2 will leave a sequence of 0s and 1s that become the binary equivalent of the decimal number. Suppose it is required to convert the decimal number N into binary form, dividing N by 2 in the decimal system, we will obtain a quotient N1 and a remainder R1, where R1 can have a value of either 0 or 1. The process is repeated until the quotient becomes 0 or 1. When the quotient is '0' or '1', it is the final remainder value. Write the final answer starting from final remainder value obtained to the first remainder value obtained.

**Example**

Convert $(65)_{10}$ into its equivalent binary number



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